Modular vertical farm

EMAE 260 · team project

Engineering problem and work

Designed a modular five-trough, 25-plant concept for a parking-garage wall. Contributed SolidWorks frame modeling, structural loading and FEA assessment, material choices and bill-of-materials trade-offs.

Process

Constraints → concepts → CAD → load cases → FEA

Tools

SolidWorks · FEA · BOM · Structural loads

Scope and status

6 ft 8 in × 8 ft × 2 ft is the documented design envelope

Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/document/d/1Qyvjt90qusbORN-KW_JqjrOvnCvlW-SjXWdgH_4e5kA/edit